

JONATHAN LYASHKO

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering

Sep. 2023 – May 2028

Waterloo, Ontario

EXPERIENCE

Research Assistant

Jan. 2026 – Apr. 2026

IDEAs Clinic, University of Waterloo

Ontario

- Led development of a **centralized robot arbitration layer** for multi-agent Tetrix robot fleets, coordinating telemetry ingestion, task allocation, and live operator control through a **multithreaded Python TCP server**.
- Built fault-tolerant **serial communication** between FPGA modules and resource-constrained microcontrollers, optimizing data transfer under strict memory, timing, and buffering constraints.
- Developed odometry-driven coordination logic for autonomous robot fleets, implementing collision avoidance and **shared-environment** navigation across multiple physical agents.
- Implemented a real-time **FFT audio visualization system** on an Intel Max 10 FPGA using **SystemVerilog**, streaming spectral analysis results to a VGA display.

AI/ML Engineer Intern

May 2025 – Aug. 2025

Revvity

Ontario

- Created **retrieval-augmented generation (RAG)** pipelines and prompt-engineered LLM behavior for domain-specific contexts, boosting output accuracy and adoption.
- Built modular, agentic LLM workflows using **LangGraph**, deploying across AWS Bedrock and Azure AI Agents to support scalable, **cloud-agnostic AI service orchestration**.
- Designed and deployed **multi-tenant AI infrastructure** with Docker, GitHub Actions, and Terraform to serve Claude 4 Sonnet, ensuring secure data partitioning and scalable enterprise access.
- Built a **seamless rollback mechanism**, transforming recovery from a high-risk, manual process into a smooth operation—dramatically minimizing downtime and business risk.
- Implemented a comprehensive **automated testing** framework in Playwright spanning **10+ system components**, proactively catching regressions before production and significantly improving reliability.

Systems Developer

Sept. 2024 – Dec. 2024

Co-operators

Ontario

- Automated workflows by developing and deploying **Python and Bash scripts** for tracking and management of areas such as SSL certificate renewal, improving efficiency and reducing manual oversight.
- Debugged and **optimized CI/CD tools**, including Dockerfiles and Jenkins build jobs, ensuring reliability through Github checks and various unit tests.
- Designed and implemented new CI/CD **pipelines using Terraform**, enhancing deployment automation and adhering the Infrastructure as Code best practices.
- Created GitHub automations with **GitHub Actions**, streamlining repository management and integration.
- Developed and maintained **.NET Web APIs using C#**, integrating internal NuGet packages to enhance functionality.

CERTIFICATIONS

AWS Certified Developer - Associate

Apr. 2024

Amazon Web Services

Microsoft Certified - Azure AI Fundamentals

Feb. 2024

Microsoft

TECHNICAL SKILLS

Programming Languages: Python, Typescript / Javascript, Rust, C++, Terraform

Developer Tools: Git, Github Actions, Postman, Jupyter Notebooks

Technologies/Frameworks: NodeJS, NextJS / React, .NET, Tensorflow, LangGraph

DevOps: Docker, Kubernetes, Helm, Vault, Linux, Jenkins, Splunk, Kong